

Set covering

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Solution of the practice quiz

The set $\mathcal{U}$ of elements is the set of programming languages:

$$\mathcal{U} = \{\text{C, C++}, \text{Java}, \text{Python}, \text{Ruby}\}.$$

Each programmer $i = 1, \dots, 4$ is associated with a subset S_i of $\mathcal{U}$, which corresponds to the known programming languages. Thus, in our case we have $n = 4$ subsets:

- $S_1 = \{\text{C, C++}\},$
- $S_2 = \{\text{Java, C++}\},$
- $S_3 = \{\text{Ruby, C++}, \text{Python}\},$
- $S_4 = \{\text{C, Java}\}.$

They are characterized by binary parameters a_{ij} that is 1 if language j is known by person i , and 0 otherwise.

$a_{1,\text{C}} = 1,$	$a_{2,\text{C}} = 0,$	$a_{3,\text{C}} = 0,$	$a_{4,\text{C}} = 1,$
$a_{1,\text{C++}} = 1,$	$a_{2,\text{C++}} = 1,$	$a_{3,\text{C++}} = 1,$	$a_{4,\text{C++}} = 0,$
$a_{1,\text{Java}} = 0,$	$a_{2,\text{Java}} = 1,$	$a_{3,\text{Java}} = 0,$	$a_{4,\text{Java}} = 1,$
$a_{1,\text{Python}} = 0,$	$a_{2,\text{Python}} = 0,$	$a_{3,\text{Python}} = 1,$	$a_{4,\text{Python}} = 0,$
$a_{1,\text{Ruby}} = 0,$	$a_{2,\text{Ruby}} = 0,$	$a_{3,\text{Ruby}} = 1,$	$a_{4,\text{Ruby}} = 0.$

The decision variables are x_1, x_2, x_3, x_4 . They are binary variables representing whether person i is involved in the team. For each language, we need

a constraint that requires that at least one person masters it. For example, for C we have

$$\sum_{i=1}^4 a_{i,C} x_i \geq 1,$$

$$x_1 + x_4 \geq 1.$$

The set of covering constraints is therefore:

$$\begin{aligned} x_1 + x_4 &\geq 1, \\ x_1 + x_2 + x_3 &\geq 1, \\ x_2 + x_4 &\geq 1, \\ x_3 &\geq 1, \\ x_3 &\geq 1. \end{aligned}$$

As we want to minimize the number of subsets, the costs c_i of each subset must all have the same value, say 1. The optimization problem is therefore

$$\min_{x \in \{0,1\}^4} x_1 + x_2 + x_3 + x_4,$$

subject to

$$\begin{aligned} x_1 + x_4 &\geq 1, \\ x_1 + x_2 + x_3 &\geq 1, \\ x_2 + x_4 &\geq 1, \\ x_3 &\geq 1, \\ x_3 &\geq 1, \\ x_i &\in \{0,1\}, \quad i = 1, \dots, 4. \end{aligned}$$

It is clear that x_3 must be included, as Tim is the only one who can cover Python and Ruby. It is also clear that at least another person must be hired, as Tim does not know C and Java. These two languages can be covered by Donald, so that we cover all languages with two persons. The optimal solution is therefore:

$$x_1 = 0, x_2 = 0, x_3 = 1, x_4 = 1.$$

The total cost is 2, and the constraints are satisfied:

$$x_1 + x_4 = 1 \geq 1,$$

$$x_1 + x_2 + x_3 = 1 \geq 1,$$

$$x_2 + x_4 = 1 \geq 1,$$

$$x_3 = 1 \geq 1,$$

$$x_3 = 1 \geq 1.$$